Semestertoets 1 1ste semester 2003 / Semester test 1 1st semester 2003

Vraag 1 / *Question* 1

Los op: / *Solve:* (a) $e^{4x} = 12$

Antw / *Ans*: $\ln(e^{4x}) = \ln(12)$ $\therefore\ 4x = \ln 12$ $\therefore\ x = \frac{\ln 12}{4} \approx 0.62123$

(b) $\frac{1}{2}\ln x = 1 - \ln 2$

Antw / *Ans*: $\ln x = 2(\ln e - \ln 2) = 2\ln\frac{e}{2}$ $\therefore\ \ln x = \ln\left(\frac{e}{2}\right)^2$

$\therefore\ x = \left(\frac{e}{2}\right)^2 \approx 1.8473$

[2]

Vraag 2 / *Question* 2

Bepaal (indien moontlik) die volgende limiete:

Find (if possible) the following limits:

(a) $\lim\limits_{x\to 2}\frac{\pi}{2}$ Antw / *Ans*: $\lim\limits_{x\to 2}\frac{\pi}{2} = \frac{\pi}{2}$

(b) $\lim\limits_{x\to\frac{\pi}{2}}\tan x$ Antw / *Ans*: $\lim\limits_{x\to\frac{\pi}{2}}\tan x \nexists$ want /*since* $\lim\limits_{x\to\frac{\pi}{2}^-}\tan x = \infty$

$\neq \lim\limits_{x\to\frac{\pi}{2}^+}\tan x = -\infty$

(c) $\lim\limits_{x\to 1^+}\frac{x^2-3x+2}{x^2-2x+1}$ Antw / *Ans*: $\lim\limits_{x\to 1^+}\frac{x^2-3x+2}{x^2-2x+1}$ $\left(\begin{matrix}\text{Vorm / } \textit{Form} \\ \frac{0}{0}\end{matrix}\right)$

$= \lim\limits_{x\to 1^+}\frac{(x-1)(x-2)}{(x-1)(x-1)} = \lim\limits_{x\to 1^+}\frac{(x-2)}{(x-1)} = -\infty$

[3]

Vraag 3 / *Question* 3

Bepaal (indien moontlik) / *Find (if possible)* $\lim\limits_{x\to-\infty}\frac{|x-1|}{x}$.

Antw / *Ans*: $\lim\limits_{x\to-\infty}\frac{|x-1|}{x} = \lim\limits_{x\to-\infty}\frac{-(x-1)}{x} = \lim\limits_{x\to-\infty}\frac{x(-1+\frac{1}{x})}{x} = \lim\limits_{x\to-\infty}(-1+\frac{1}{x}) = -1$

[2]

Vraag 4 / *Question* 4

Laat $F(x) = \sin^2 3x$ en $f(x) = \sin 3x$. Gee 'n funksie g sodat $F = g\circ f$.

Let $F(x) = \sin^2 3x$ *and* $f(x) = \sin 3x$. *Give a function* g *such that* $F = g\circ f$.

Antw / *Ans*: Laat / *Let* $g(x) = x^2$

dan / *then* $F(x) = (g\circ f)(x) = g(f(x)) = g(\sin 3x) = (\sin 3x)^2$

[1]

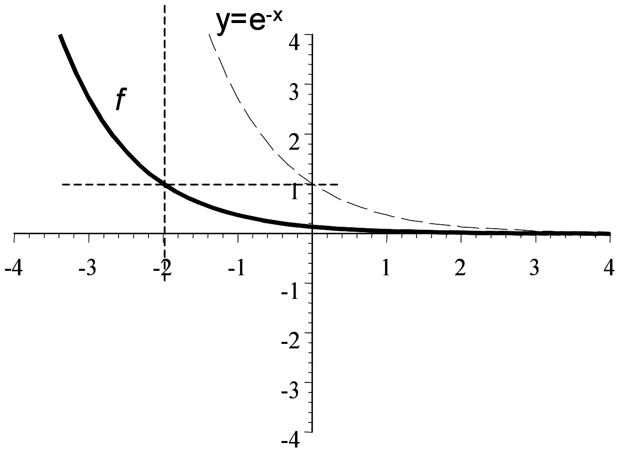
Vraag 5 / *Question* 5

Skets die volgende funksies op die gegewe assestelsels:

Sketch the following functions on the given sets of axes:

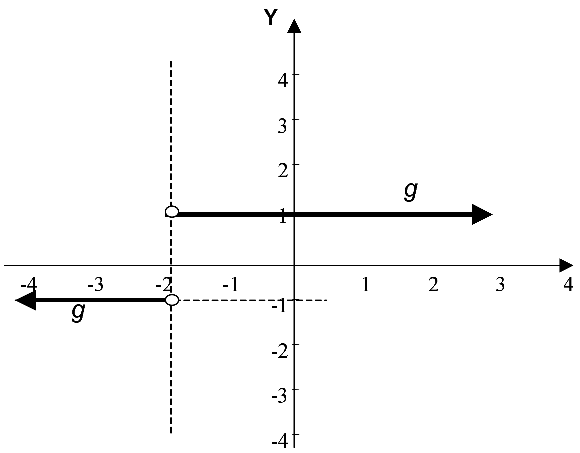
(5.1) $f(x) = e^{-x-2}$

Antw / *Ans*: $f(x) = e^{-x-2} = e^{-(x+2)}$ ($y = e^{-x}$ *shifted* 2 *units to the left*)

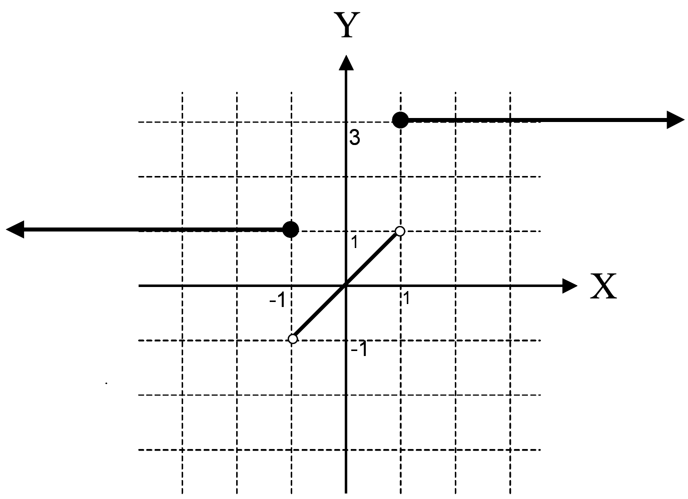


(5.2) Antw / *Ans*: $$g(x) = \frac{x+2}{|x+2|} = \begin{cases} \dfrac{x+2}{(x+2)} & \text{as / } if\ x+2>0 \\ \dfrac{x+2}{-(x+2)} & \text{as / } if\ x+2<0 \end{cases}$$

$$= \begin{cases} 1 & \text{as / } if\ x > -2 \\ -1 & \text{as / } if\ x < -2 \end{cases}$$

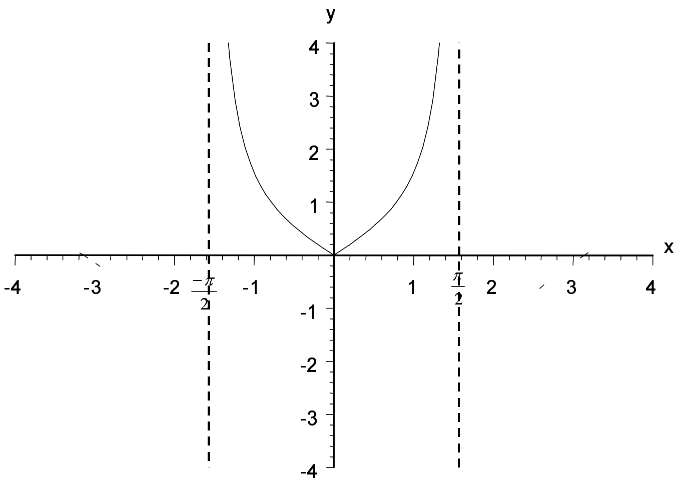


(5.3) $$h(x) = \begin{cases} 1 & \text{as / } if\ x \le -1 \\ x & \text{as / } if\ |x| < 1 \\ 3 & \text{as / } if\ x \ge 1 \end{cases} = \begin{cases} 1 & \text{as / } if\ x \le -1 \\ x & \text{as / } if\ -1 < x < 1 \\ 3 & \text{as / } if\ x \ge 1 \end{cases}$$



(5.4) $F(x) = \tan|x|, \quad x \in \left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$

$$= \begin{cases} \tan(x) & \text{as / } \textit{if } x \geq 0 \\ \tan(-x) & \text{as / } \textit{if } x < 0 \end{cases} = \begin{cases} \tan x & \text{as / } \textit{if } x \geq 0 \\ -\tan x & \text{as / } \textit{if } x < 0 \end{cases}$$



[6]

Vraag 6 / *Question* 6

Vind die definisie- en waardeversameling van $f(x) = \sin^{-1}(x-2)$.

Find the domain and range of $f(x) = \sin^{-1}(x-2)$.

(Notasie / *Notation:* $\sin^{-1}x = \arcsin x = bg\sin x$.)

Antw / *Ans*: $f(x) = \sin^{-1}(x-2)$
$\Leftrightarrow \sin f(x) = x-2$ met / *with* $-1 \leq x-2 \leq 1$ en / *and* $f(x) \in [-\frac{\pi}{2}, \frac{\pi}{2}]$.
$\therefore$ definisieversameling / *domain is* $D_f = [1,3]$
want / *since* $-1+2 \leq x-2+2 \leq 1+2$
$1 \leq x \leq 3$
waardeversameling / *range is* $W_f = [-\frac{\pi}{2}, \frac{\pi}{2}]$

[2]

Vraag 7 / *Question* 7

Los op vir x as $\sin^{-1}\sqrt{2x} = \cos^{-1}\sqrt{x}$.

Solve for x *if* $\sin^{-1}\sqrt{2x} = \cos^{-1}\sqrt{x}$.

(Notasie / *Notation:* $\sin^{-1}x = \arcsin x = bg\sin x$.)

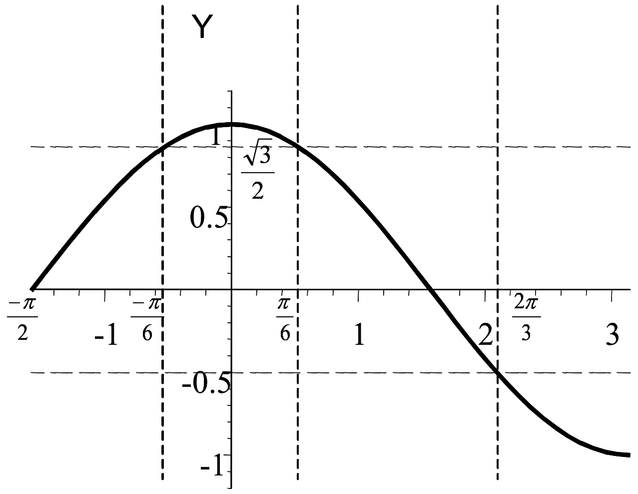
Antw / *Ans*: Laat / Let $\sin^{-1}\sqrt{2x} = y = \cos^{-1}\sqrt{x}$ dan / *then*: $\sin y = \sqrt{2x}$ en / *and* $\cos y = \sqrt{x}$.
maar / *but* $\sin^2 y + \cos^2 y = 1$ $\therefore$ $\left(\sqrt{2x}\right)^2 + (\sqrt{x})^2 = 1$ $\therefore$ $2x + x = 1$ $\therefore$ $x = \frac{1}{3}$
want / *since* $y \in [-\frac{\pi}{2}, \frac{\pi}{2}] \cap [0, \pi] = [0, \frac{\pi}{2}]$

[2]

Vraag 8 / *Question* 8

Los op: / *Solve*: $-1 \leq 2\cos x \leq \sqrt{3}$, $x \in \left[\frac{-\pi}{2}, \pi\right]$

Antw / *Ans*: $\frac{-1}{2} \leq \cos x \leq \frac{\sqrt{3}}{2}$ $x \in [\frac{-\pi}{2}, \frac{-\pi}{6}] \cup [\frac{\pi}{6}, \frac{2\pi}{3}]$ want / *since*



[3]

<u>Vraag 9 / *Question* 9</u>

Laat / *Let* $\quad f(x) = \begin{cases} ax, & x \le 2 \\ 2x+1, & x > 2 \end{cases}$.

(9.1) Bepaal a sodat f oral kontinu is.

Determine a such that f is continuous everywhere.

Antw / *Ans*: ax en $2x+1$ is polinome wat oral kontinu is, behalwe dalk in $x = 2$ waar die funksies ontmoet.

ax and $2x+1$ are polynomials that are continuous everywhere but perhaps not where the two functions meet at $x = 2$.

$f(2) = 2a$

$\lim_{x\to 2^-} f(x) = \lim_{x\to 2^-} (ax) = 2a = \lim_{x\to 2^+} f(x) = \lim_{x\to 2^-} (2x+1) = 5$

$\Leftrightarrow 2a = 5 \qquad \therefore\ a = \frac{5}{2}$

[2]

(9.2) Is f differensieerbaar vir hierdie a? Verduidelik.

Is f differentiable for this a? Explain.

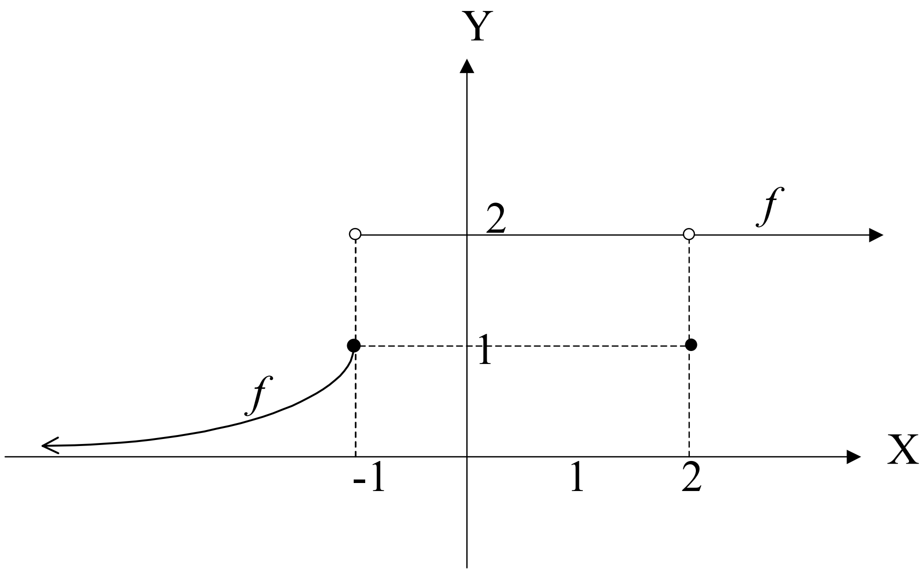
Antw / *Ans*: $\quad f(x) = \begin{cases} \frac{5}{2}x, & x \le 2 \\ 2x+1, & x > 2 \end{cases} \qquad f'_l(2) = \frac{5}{2} \neq 2 = f'_R(2). \qquad \therefore\ f$ *is not differentiable at* $x = 2$.

[2]

<u>Vraag 10 / *Question* 10</u>

Die grafiek van f word gegee.

The graph of f is given.



(10.1) Sê of die limiet bestaan en gee sy waarde indien wel.
State if the limit exists and if so, give its value.

(a) $\lim\limits_{x \to -1^-} f(x)$ Antw / *Ans*: $\lim\limits_{x \to -1^-} f(x) = 1$

(b) $\lim\limits_{x \to -1} f(x)$ Antw / *Ans*: $\lim\limits_{x \to -1^+} f(x) = 2 \neq \lim\limits_{x \to -1^-} f(x)$

$\therefore \lim\limits_{x \to -1} f(x) \nexists$

(c) $\lim\limits_{x \to 1} f(x)$ Antw / *Ans*: $\lim\limits_{x \to 1} f(x) = 2$

(d) $\lim\limits_{x \to -\infty} f(x)$ Antw / *Ans*: $\lim\limits_{x \to -\infty} f(x) = 2$

[4]

(10.2) Skryf die x – *waardes* waar f nie differensieerbaar is nie, neer.

Write down the x – values where f is not differentiable.

Antw / *Ans*: $x = -1, \quad x = 2.$
want as f diskontinu is, is f nie differensieerbaar nie
/ *since if f is discontinuous then f is not differentiable.*

[2]

(10.3) Skryf die interval(le) waar f kontinu is neer.

Write down the interval(s) where f is continuous

Antw / *Ans*: $(-\infty, -1], \quad (-1, 2), \quad (2, \infty)$

[2]

Vraag 11 / *Question* 11

Laat / *Let* $f(x) = |2x - 1|$.

Gebruik die definisie van die afgeleide (m.a.w. eerste beginsels) om aan te toon dat f nie differensieerbaar is in $x = \frac{1}{2}$ nie.

Use the definition of the derivative (i.e. first principles) to show that f is not differentiable at $x = \frac{1}{2}$.

Antw / *Ans*: $f(x) = |2x - 1| = \begin{cases} 2x - 1 & \text{as / } if \ 2x - 1 \geq 0 \\ -(2x - 1) & \text{as / } if \ 2x - 1 < 0 \end{cases}$

$$= \begin{cases} 2x-1 & \text{as / if } x \geq \frac{1}{2} \\ -(2x-1) & \text{as / if } x < \frac{1}{2} \end{cases}$$

$$f_L'\left(\tfrac{1}{2}\right) = \lim_{x \to \frac{1}{2}^-} \frac{f(x)-f(\frac{1}{2})}{x-\frac{1}{2}} = \lim_{x \to \frac{1}{2}^-} \frac{-(2x-1)-0}{x-\frac{1}{2}} = \lim_{x \to \frac{1}{2}^-} \frac{-2\left(x-\frac{1}{2}\right)}{x-\frac{1}{2}} = -2$$

$$f_R'\left(\tfrac{1}{2}\right) = \lim_{x \to \frac{1}{2}^+} \frac{f(x)-f(\frac{1}{2})}{x-\frac{1}{2}} = \lim_{x \to \frac{1}{2}^+} \frac{(2x-1)-0}{x-\frac{1}{2}} = \lim_{x \to \frac{1}{2}^+} \frac{2\left(x-\frac{1}{2}\right)}{x-\frac{1}{2}} = 2 \neq f_L'\left(\tfrac{1}{2}\right)$$

$$\therefore f'\left(\tfrac{1}{2}\right) \nexists$$

alternatief / *alternative:*

$$f_L'\left(\tfrac{1}{2}\right) = \lim_{h \to 0^-} \frac{f(\frac{1}{2}+h)-f(\frac{1}{2})}{h} = \lim_{h \to 0^-} \frac{-\left(2\left(\frac{1}{2}+h\right)-1\right)-0}{h} = \lim_{h \to 0^-} \frac{-2h}{h} = -2$$

$$f_R'\left(\tfrac{1}{2}\right) = \lim_{h \to 0^+} \frac{f(\frac{1}{2}+h)-f(\frac{1}{2})}{h} = \lim_{h \to 0^+} \frac{\left(2\left(\frac{1}{2}+h\right)-1\right)-0}{h} = \lim_{h \to 0^+} \frac{2h}{h} = 2 \neq f_L'\left(\tfrac{1}{2}\right)$$

$$\therefore f'\left(\tfrac{1}{2}\right) \nexists$$

$\therefore$ f is nie differensieerbaar in $x = \frac{1}{2}$ nie.

f is not differentiable at $x = \frac{1}{2}$.

[3]